ETHNOHACK 2026 · TRADITIONAL KNOWLEDGE TRACK

EthnoVault

A consent-first living archive for Indigenous oral traditions, ethnopharmacology, and tribal languages.

10	4	10	3 files
PUBLISHED RECORDS	FEATURED RECORDINGS	REGIONAL MAP PINS	ZERO-BACKEND SPA

Live demo: hariprasath71811-max.github.io/ethnovault · Repo: github.com/hariprasath71811-max/ethnovault

THE PROBLEM

Oral knowledge disappears when it is extracted without consent.

Disappearing knowledge

UNESCO estimates a language dies every two weeks. Plant wisdom, songs, and ecological calendars live in elders' voices — not in databases.

Extractive archives

Most archives record first and ask later: no attribution, no reuse terms, no withdrawal. Communities lose control of their own heritage.

No FPIC in tooling

Free, Prior and Informed Consent is a legal principle, but almost no software enforces it at the media layer.

Verification gap

There is no lightweight way for elders and culture-bearers to review, endorse, or reject a record before it goes public.

THE SOLUTION

Ethical ethnopharmacology + an acoustic, community-governed archive.

Consent-first by design

Every record carries attribution, community, region, license tier (listen-only / educational / community-internal) and an explicit consent flag. Nothing publishes without quorum.

Acoustic Oral Studio

Waveform visualizer, scrubber, transcript-synced seeking, speed control, and a Web Audio harmonic synthesizer for language-learning playback.

Elder Council verification

A 3-endorsement quorum from elders and culture-bearers publishes a record. Communities can withdraw at any time.

Guided submission pipeline

Knowledge type, origin community, attribution, story, optional voice recording, and consent — in one flow. New records enter a pending queue.

DATASET & KNOWLEDGE SCHEMA

10 published + 2 pending records across 6 knowledge domains.

Field	Examples
id / title	sami-aurora, ayahuasca-botany, toda-embroidery, hokulea-navigation, ainu-yukar…
category	Ethnomedicine, Traditional Crafts, Oral Folklore, Ethnobotany, Ecological Knowledge, Navigation
community / region	Sámi (Nordic), Shipibo-Konibo (Peru), Toda (Nilgiris), Māori (Aotearoa), Ainu (Hokkaido)…
attribution / license	Named knowledge keepers · listen-only / educational / community-internal
media	Attributed audio, transcript (original + community-approved translation), tags, status
status	published · pending (Elder Council) · verified (3 endorsements)

Records are illustrative composites for the hackathon — not real community records. All media is simulated.

TECHNICAL ARCHITECTURE

Three static files. No backend. No tracking.

Frontend

index.html + styles.css + app.js. Tailwind CSS (CDN), Lucide icons, Google Fonts, Unsplash imagery. State persisted in localStorage.

Audio layer

Web Audio API harmonic synthesizer + Canvas waveform renderer. Real-time playhead, scrubbing, speed control, transcript-synced seeking.

Rendering

Vanilla JS SPA with hash-free in-page navigation, IntersectionObserver reveals, requestAnimationFrame counters, responsive grid (390–1440 px).

Deployment

GitHub Pages (static). Automated demo recording with Playwright + ffmpeg (H.264, 83.8 s). Zero build step, zero dependencies at runtime.

KEY FEATURES

Ethnopharmacology catalog + Acoustic Oral Studio.

Archive & search

- Live search across title, community, region, tags
- Category + region filters, map-pin filtering
- Result counts, active-filter chips, clear-all
- Narrative story modals with consent labels

Oral Studio

- Waveform with draggable playhead
- Original + community-approved translation
- Click any transcript line to seek
- Speed, volume, ARIA slider, keyboard control

Impact & map

- Animated counters (stories, languages, elders)
- Interactive regional map with pins
- Dark mode, reduced-motion support

Accessibility

- Focus traps in modals, ARIA labels
- Keyboard-operable waveform
- Zero horizontal overflow 390–1440 px

ELDER COUNCIL VERIFICATION

Nagoya Protocol & FPIC compliance, built into the workflow.

- **Quorum:** 3 endorsements from elders + culture-bearers required to publish.
- **Pending queue:** every submission starts pending; never auto-published.
- **Attribution:** knowledge keepers named on every record, with reuse terms.
- **Withdrawal:** communities can withdraw a record at any time (deletion tooling on roadmap).
- **Privacy tiers:** listen-only, educational, community-internal — enforced at the media layer.
- **Auditability:** endorsement events are visible and traceable in the council view.

3	FPIC	Nagoya	0
ENDORSEMENT QUORUM	CONSENT ENFORCED	PROTOCOL ALIGNED	AUTO-PUBLISHED RECORDS

PROTOTYPE RESULTS

Live, verified, and accessible.

Live verification

- Deployed: GitHub Pages, HTTP 200, no console errors
- Repo public on `main` (2 commits)
- Demo video: 83.8 s, H.264, 1600×900, 30 fps
- YouTube: https://youtu.be/1q7em74otS0

Quality checks

- Zero horizontal overflow 390–1440 px
- Dark mode + reduced-motion support
- Keyboard + screen-reader friendly
- Lighthouse-friendly static bundle

83.8s
DEMO DURATION

10 MB
VIDEO SIZE

3 files
RUNTIME BUNDLE

0
BACKEND SERVICES

Honest about what's simulated. Clear about what's next.

Limitations (simulated)

- Audio, uploads, translations, endorsements, council actions run in-browser
- Records are illustrative composites, not real community knowledge
- No real identity, auth, or encrypted storage yet

Responsible research

- Consent-first, attribution on every record
- Community-controlled reuse tiers
- No extraction, no tracking, no surveillance

Roadmap

Encrypted community-held media storage · Offline-first PWA for low-connectivity regions · Production council identity & audit trails · True withdrawal/deletion tooling · Time-aligned transcripts & pronunciation guides · Multi-language interfaces · Partnerships with language nests & seed keepers.